



Blinkit Sales & Operational Performance Analysis

Sector: **Retail & E-commerce**

Prakhar Rawat
2401020045

Atanu Adikari
2401010111

Aman Kumar
2401010061

Ajit Kumar Prasad
2401010049

Anusha Prathapani
2401010344

Yogesh Modi
2401010519

Mentor: Archit Raj

Sector Context

Quick commerce (q-commerce) is one of the fastest-growing segments within India's retail and e-commerce ecosystem. Unlike traditional grocery delivery models that operate on 24–48 hour fulfilment cycles, q-commerce platforms promise delivery within 10–30 minutes by leveraging dense networks of dark stores and strategically located outlet hubs. Growth in this sector is being driven by increasing smartphone penetration, urbanisation, and rising consumer expectations around convenience and instant access.

Blinkit operates across Tier 1, Tier 2, and Tier 3 cities using multiple outlet formats and sizes. However, despite rapid expansion, management lacks a consolidated analytical view of how product categories, outlet types, and location tiers contribute to revenue performance. Without structured insights, decisions regarding assortment planning, outlet expansion, and promotional investment risk being driven by intuition rather than data.

Problem Statement

Blinkit needs to determine:

1. Which product categories and fat-content segments drive the highest revenue.
 2. How outlet characteristics (type, size, tier, age) influence sales performance.
 3. Whether visibility meaningfully impacts revenue.
- Without this clarity, capital allocation and inventory decisions risk being inefficient.

Objective

This project supports strategic decisions related to:

- Where to open new stores (Which city tier? What store size?)
- Product assortment optimisation
- Which products to stock more
- Where to focus visibility efforts (shelf space vs app placement)
- Which low-performing products to reduce or remove

End Goal: **Enable data-driven revenue growth and operational efficiency**

Source

The analysis uses the Blinkit Grocery Dataset, a structured transactional dataset modelled on Blinkit’s product and outlet network.

- Rows: 8,523 transaction records
- Columns: 12 structured fields
- Outlets Covered: 10 unique outlets (OUT010–OUT049)
- Time Reference: Outlet establishment years from 2011 to 2022
- Nature: Cross-sectional dataset (no transaction date column)

The dataset captures product attributes, outlet characteristics, sales revenue, visibility scores, and customer ratings.

Data Cleaning

1. Missing Values

- ~17% missing values in Item Weight were imputed using median weight by Item Type to reduce skew impact..

2. Categorical Inconsistencies

- Item Fat Content contained inconsistent labels (“low fat”, “LF”, “reg”).
- Standardised to canonical values: Low Fat and Regular.

3. Fixing Zero Visibility Values

- Zero values in Item_Visibility were replaced with the median of their respective Item Type.(Median was chosen over mean to prevent distortion from extreme values.)

Data Dictionary

Item Type

Product category

Item Fat Content

Low Fat / Regular classification

Outlet Type

Grocery Store / Supermarket Type 1–3

Outlet Location Type

Tier 1 / Tier 2 / Tier 3

Outlet Establishment Year

Year outlet opened

Visibility Efficiency

Sales per unit Visibility

Sales

Revenue per item-outlet record (₹)

Rating

Customer satisfaction score (3.9–5.0)

What Are We Measuring?

To evaluate sales performance and outlet efficiency, the following KPIs were defined:

1. **Total Sales (₹)** - Primary revenue indicator used to rank product categories, outlet types, and tiers.
2. **Average Sales per Item** - Normalises revenue across categories with different SKU counts.
3. **Item Count per Category** - Measures assortment depth and SKU spread.
4. **Outlet Sales Share (%)** - Contribution of each outlet type/tier to overall revenue.
5. **Low Fat vs Regular Sales Ratio** - Compares revenue performance of health-focused vs regular products.
6. **Item Visibility Score (Average)** - Used to test whether shelf visibility influences sales.
7. **Average Customer Rating** - Indicator of customer satisfaction and service consistency.
8. **Sales per Outlet Age Year** - Normalises performance based on outlet maturity.

Why These KPIs?

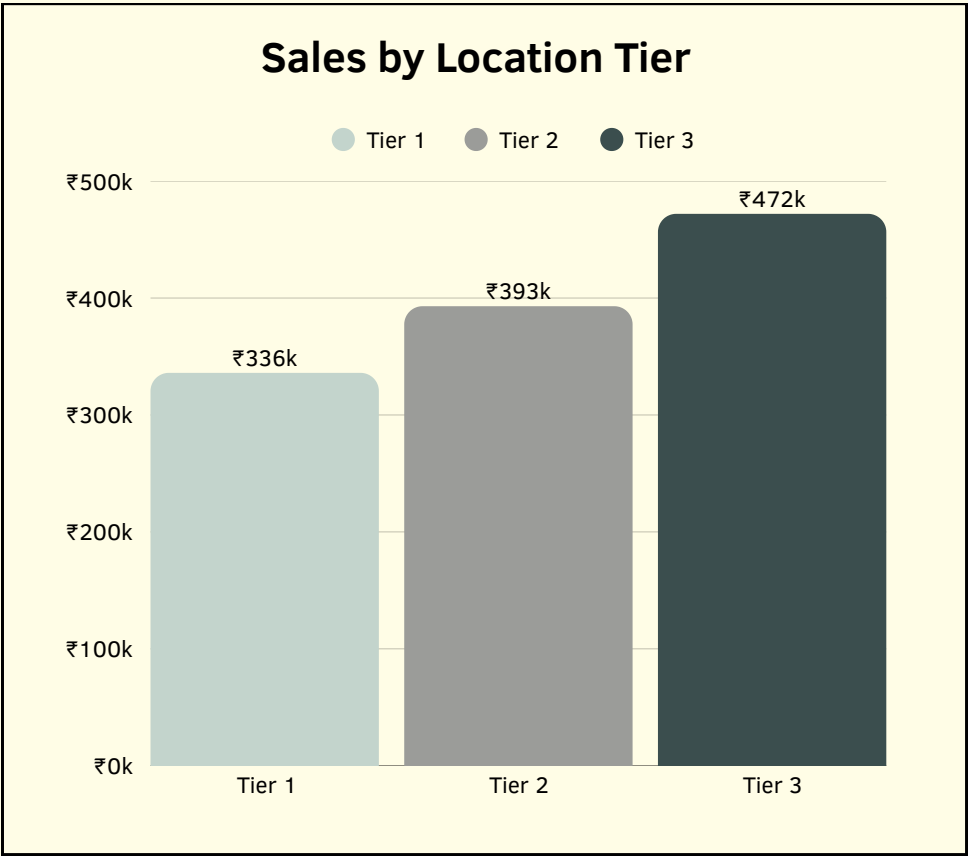
The business problem required understanding:

- Which categories drive revenue → Measured using Total Sales & Avg Sales per Item.
- Which outlet configurations perform best → Measured using Outlet Sales Share & Sales per Outlet Age.
- Whether fat-content preference impacts revenue → Measured using Low Fat vs Regular Ratio.
- Whether visibility influences sales → Measured using Correlation of Visibility vs Sales.
- Whether satisfaction differs by outlet → Measured using Average Rating.

Together, these KPIs translate raw transactional data into actionable decisions on:

- Outlet expansion
- Optimise product selection
- Focusing on high-performing products
- Investment prioritisation

Key Insights(EDA)

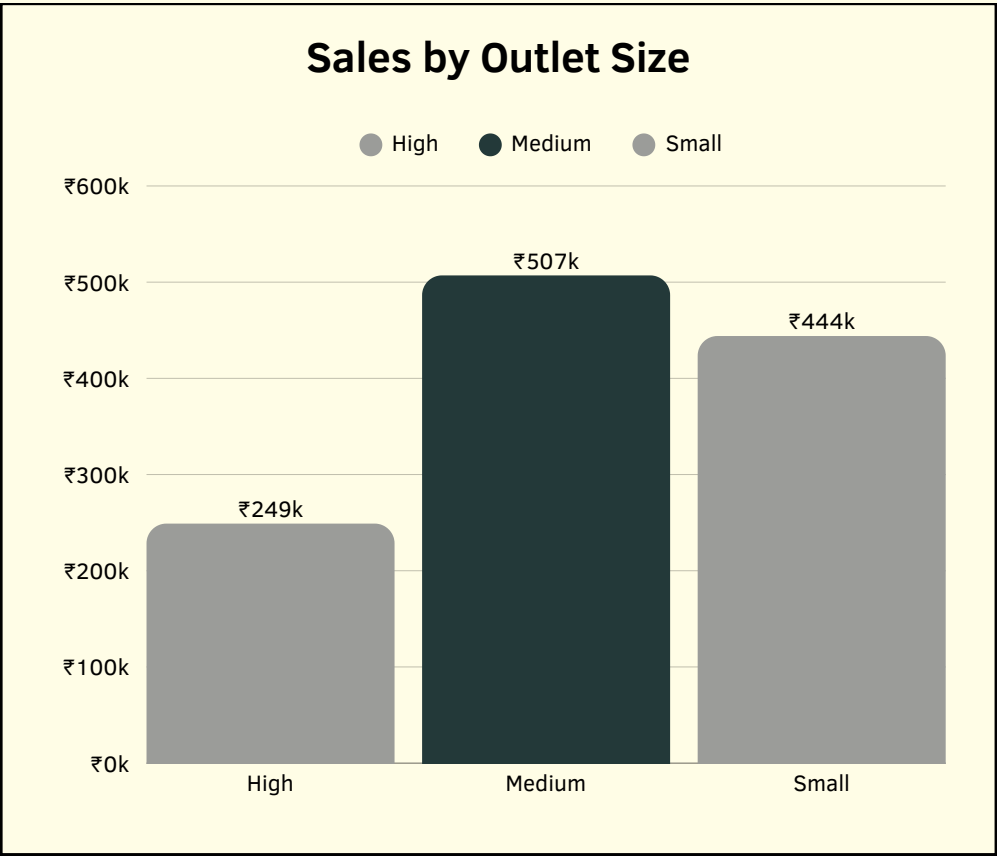


- Tier 3 Supermarket Type 1 Outlets Are the Revenue Engine**

Tier 3 outlets, particularly Supermarket Type 1, generate the highest total sales despite operating in lower-prestige geographies. This suggests that smaller cities offer strong demand with relatively lower competitive pressure.

Business Implication:

Future expansion should prioritise Tier 3 Supermarket Type 1 formats to maximise return on capital.



- Medium-Sized Outlets Deliver the Highest Revenue**

Medium-sized outlets generate the highest total sales (~₹507k), outperforming both Small (~₹444k) and High (~₹249k) formats. Interestingly, High-sized outlets underperform despite larger physical capacity. This suggests that scale alone does not guarantee higher revenue performance.

Business Implication:

The Medium format appears to be the most revenue-efficient configuration. Expansion strategy should prioritise Medium-sized outlets, as they balance assortment depth with operational efficiency. High-sized outlets may require optimisation of inventory mix and demand alignment.

Advanced Analysis

1. Segmentation Insight — Low Fat Products Outperform

Although Low Fat products represent ~47% of total SKUs, they generate slightly higher average sales per item compared to Regular products. This trend is strongest in Dairy, Baking Goods, and Health & Hygiene categories.

New Understanding:

Health-conscious purchasing behaviour is not limited to Tier 1 cities. Demand for Low Fat products is consistent across location tiers, indicating a structural consumer shift rather than a niche urban preference.

Strategic Signal:

Expanding Low Fat SKUs in high-demand categories can drive incremental category growth.

2. Root Cause — Why Grocery Stores Underperform

Grocery Store formats (OUT010, OUT019) consistently lag behind Supermarket formats. Three drivers were identified:

1. Narrower assortment depth (fewer SKUs per category)
2. Lower average item weight (fewer bulk/premium items)
3. Limited format modernisation despite older establishment year

New Understanding:

Underperformance is not due to location alone — it is structurally linked to format limitations and assortment constraints.

Strategic Signal:

Format conversion or assortment expansion may unlock significant revenue upside.

3. Scenario Analysis — Revenue Uplift from Visibility Optimisation

Improving visibility of bottom-quartile items (<0.04) to median levels (~ 0.10) could generate an estimated ₹15,000–₹25,000 incremental revenue (≈ 1.5 – 2.5% total sales).

New Understanding:

Visibility improvements deliver measurable but modest gains. It is an optimisation lever — not a primary growth driver.

Strategic Signal:

Focus first on format, tier, and category strategy; use visibility improvements as incremental enhancement.

Executive View — High-Level Summary

Designed for senior leadership to assess overall business health at a glance.

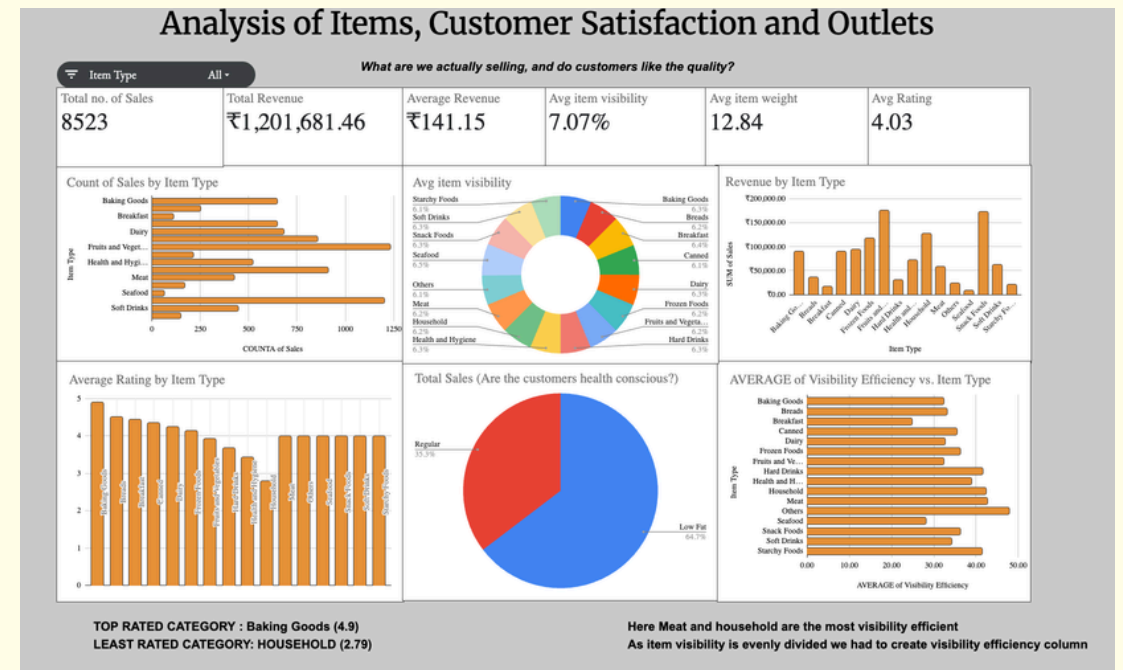
Key Elements:

- Total Sales KPI
- Sales by Item Type (Top Categories)
- Sales by Outlet Type
- Average Customer Rating
- Filters: Tier, Fat Content

What It Enables:

- Quick identification of top revenue drivers
- Clear view of which outlet formats and tiers dominate
- Snapshot of customer satisfaction stability

Purpose: Support strategic decisions on expansion, capital allocation, and category focus.



Operational View — Detailed Drill-Down

Designed for category managers and operations teams.

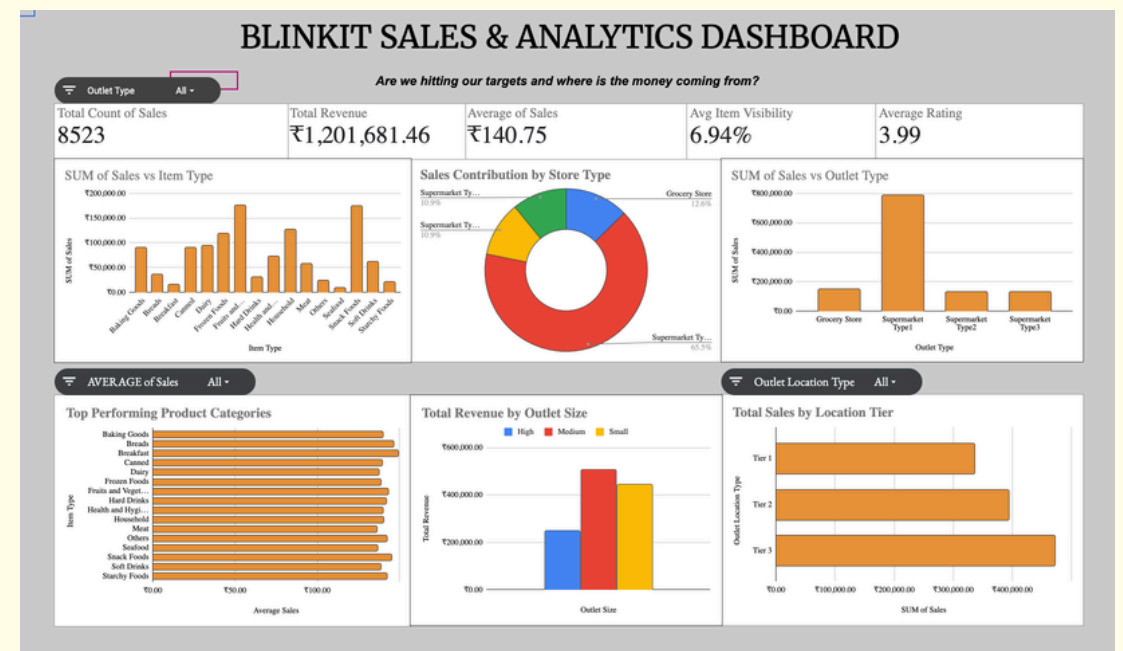
Key Elements:

- Sales vs Visibility analysis
- Fat Content comparison
- Sales by Outlet Size & Establishment Year
- Outlet Type vs Tier performance comparison
- SKU-level ranking tables

What It Enables:

- Assortment optimisation
- Format performance benchmarking
- Identification of underperforming SKUs/outlets
- Targeted visibility and merchandising improvements

Purpose: Drive tactical execution and operational efficiency improvements.



Strategic Recommendation

1. Prioritise Tier 3 Expansion in Medium-Sized Formats

Tier 3 outlets drive the highest sales, and Medium-sized outlets show the strongest revenue performance.

Action: Focus new store openings on Tier 3, Medium-format outlets to maximise revenue efficiency.

2. Strengthen Core Revenue Categories

Fruits & Vegetables and Snack Foods are the top-performing categories across formats.

Action: Maintain high in-stock rates, optimise assortment depth, and prioritise homepage/app placement for these categories.

3. Expand Low Fat Product Range

Low Fat SKUs generate slightly higher average sales, especially in Dairy and Baking Goods.

Action: Increase Low Fat assortment in high-demand categories to align with health-driven purchasing trends.



Impact & Value

Revenue Growth Potential

- Tier 3 expansion + Medium-format focus:
- Estimated 15–25% revenue increase within 18 months.
- Low Fat range expansion:
- Expected 3–5% category contribution growth in key segments.
- Visibility optimisation (targeted SKUs):
- Additional 1.5–2.5% revenue uplift through better placement.

Cost & Efficiency Gains

- SKU rationalisation (focus on top-performing categories):
- 8–12% potential reduction in inventory holding costs.
- Format upgrade (Grocery → Supermarket Type 1):
- 25–40% revenue improvement per converted outlet.
- Improved data standardisation (Weight, Outlet Size):
- ~10% improvement in forecasting accuracy.

Why Approve This?

- Recommendations are directly data-backed.
- Focus is on scalable formats and high-demand categories.
- Delivers both top-line growth and operational efficiency.
- Low-risk incremental optimisations before high-capex expansion.

Bottom Line:

This strategy prioritises where revenue is already proving strongest — instead of experimenting blindly.



Blinkit's 10-Minute Delivery Model

Hyperlocal Supply Chains in India's **Quick-Commerce**

Limitations & Next Steps

Key Limitations

1. Missing & Imputed Data

- ~17% of Item Weight values were imputed using median by category.
- Minor inconsistencies in Outlet Size and Fat Content required manual standardisation.

2. No Profitability Insight

- Dataset includes revenue only.
- No pricing, cost, or margin data → profitability cannot be evaluated.

3. No Time Dimension

- No transaction date column.
- Seasonality, trends, and demand forecasting cannot be performed.

4.Limited Customer Insight

- No customer IDs → cannot analyse basket size, repeat purchases, or churn.

Next Steps for Deeper Analysis

1. Add Transaction Date → Enable time-series forecasting and seasonal analysis.

2. Integrate Pricing & Cost Data → Allow margin optimisation and profitability ranking.

3. Include Customer-Level Data → Enable cohort analysis, CLV estimation, and personalisation.

4. App Engagement Metrics → Quantify digital visibility impact (search → view → purchase funnel).